

Learning on the move: Designing Adaptive Education for Automated Vehicles

Kevin Fred Mwaita
Free University of Bozen-Bolzano
Bolzano, Italy
Bern University of Applied Sciences
Bern, Switzerland
kevin.mwaita@bfh.ch

Mark Colley
UCL Interaction Centre
London, United Kingdom
m.colley@ucl.ac.uk

Thiemo Wambsganss
Bern University of Applied Sciences
Bern, Switzerland
thiemo.wambsganss@bfh.ch

Abstract

Automated vehicles (AVs) could transform commute time into learning opportunities by freeing passengers from driving tasks. We conducted semi-structured interviews with 31 regular commuters, including educators and automotive experts, to identify design challenges and opportunities for AV-based learning. Using general inductive thematic analysis, we identified three key challenges: managing cognitive load and distraction, technology integration and privacy, and content design and interdisciplinary alignment. We also identified four autonomous vehicle-specific opportunities: journey-linked contextual learning, productive time utilization, controlled sensory environments, and automation-aware personalization. Our findings indicate that AV affordances, including real-time environmental data, private space, and extended uninterrupted time, create design requirements distinct from general mobile learning. We contribute design recommendations for adaptive focus management, context-aware content delivery, multi-modal interaction, and privacy-preserving personalization to guide future AV learning systems.

CCS Concepts

• **Applied computing** → *E-learning*; Interactive learning environments; • **Human-centered computing** → **Empirical studies in HCI**.

Keywords

Context-Aware Learning, Multimodal Interaction, Adaptive Learning Systems, Automated Vehicles

ACM Reference Format:

Kevin Fred Mwaita, Mark Colley, and Thiemo Wambsganss. 2026. Learning on the move: Designing Adaptive Education for Automated Vehicles. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3772363.3798922>

1 Introduction

Commuters in major cities lose up to 101 hours annually in traffic [6]. Automated vehicles (AVs) at SAE levels 4 and 5 [16] could

convert this time into learning opportunities by removing the driving task [2]. Prior work has explored non-driving-related activities (NDRAs) in AVs [13, 21], and mobile learning has been studied in public transit [5]. However, AVs present affordances that differ from other mobile contexts and warrant dedicated investigation.

AVs offer a private, controllable space where learners can customize lighting, audio, and display. AVs also provide uninterrupted time blocks, real-time environmental data from vehicle sensors that can link content to passing landmarks, and active noise control systems that reduce cabin noise below 500 Hz [17]. These features make audio-based learning more feasible than in public transit. However, motion sickness can limit screen use [3], and cabin space restricts movement-based activities.

The World Economic Forum projects that 600 million people will require upskilling by 2030 due to AI automation [22]. AVs could serve as supplementary learning spaces to address this need. Designing learning experiences for this context requires understanding stakeholder perspectives across disciplines.

We conducted an exploratory interview study with 31 regular commuters, including educators (n=9) and automotive experts (n=5), to understand challenges and opportunities for AV-based learning. Our analysis addressed three research questions:

RQ1: How do commuters with different expertise envision learning in AVs?

RQ2: What challenges and opportunities emerge when implementing education in mobile AV environments?

RQ3: What design recommendations can guide AV learning systems?

Contributions Statement: This paper contributes (1) empirical insights from diverse stakeholders on AV learning, (2) three challenges specific to AV learning contexts, (3) four AV-specific opportunities not addressed in prior mobile learning literature, and (4) design recommendations mapping AV affordances to pedagogical strategies.

2 Related Work

We situate our work at the intersection of three research areas: NDRAs in AVs, mobile and commute-based learning, and adaptive educational technologies.

2.1 Non-Driving Activities in Automated Vehicles

Research on NDRAs in AVs has examined what passengers might do when freed from driving. Wilson et al. [21] found that among likely AV adopters, preferences for leisure (88%), resting (75%), socializing



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798922>

(70%), and productive activities (55%) were the most commonly mentioned in-vehicle uses of time. Pfleging et al. [13] identified a broad set of NDRAs during automated driving, spanning communication, entertainment, productivity, and relaxation. Pudane et al. [15] explored how AVs might reshape daily activities and found that commuters envision using travel time for work and personal development. Even in non-automated contexts, Martelaro et al. [10] demonstrated that speech-based productivity tasks are feasible during driving, suggesting further potential when driving demands are removed. These studies establish that learning is among the activities commuters would consider. However, none systematically examined educational design requirements for AV contexts.

2.2 Mobile and Commute-Based Learning

Mobile learning research has examined educational use of portable devices [5]. Commute-specific learning remains understudied. Danna et al. [1] reviewed literature across transportation, education, and organizational psychology, identifying that commute learning is constrained by attention demands, motion sickness, and variable connectivity. Situated learning theory [9] suggests that authentic contexts enhance learning but also emphasizes social interaction, which is limited in individual AV commutes. Multimedia learning theory [11] provides principles for combining visual and auditory content. These principles are relevant to designing AV interfaces that minimize cognitive overload while supporting retention.

2.3 Adaptive Educational Technologies

Intelligent tutoring systems (ITS) are nearly as effective as human tutors in adapting to learner needs [8, 19]. Plass and Pawar [14] developed a taxonomy of adaptivity for learning that includes adapting content (difficulty and conceptual progression), sequence (rehearsal schedule and cross-course progression), and interaction type. These distinctions are relevant to AV contexts where environmental conditions change during a journey. Recent work on large language models suggests possibilities for conversational tutoring [7], though their application in mobile environments remains unexplored. Adaptive learning has been studied in classrooms [23] and online settings [20], but AV-specific design opportunities have been understudied.

3 Method

We conducted semi-structured interviews with regular commuters to understand their perspectives on learning in AV environments. Our methodology followed five phases: (1) literature review on NDRAs, mobile learning, and adaptive technologies to identify key topics for interview questions, (2) semi-structured interviews with video elicitation, (3) transcription and verification, (4) systematic coding using Atlas.ti, and (5) thematic analysis to derive design insights.

3.1 Participants

We recruited 31 participants through Prolific and university mailing lists over four weeks. The sample included 16 female (51.6%) and 15 male (48.4%) participants. The age distribution was as follows: 25 to 34 years (48.3%), over 55 years (22.6%), 45 to 54 years (9.7%), 35 to 44

years (12.9%), and 18 to 24 years (6.5%). Professional backgrounds included educators (n=9), automotive technology professionals (n=5), and participants from other fields including healthcare, marketing, administration, and retired individuals (n=17). Education levels included PhD or PhD student (19.4%), Master's degree (22.6%), Bachelor's degree (35.5%), and Associate degree or some college (22.6%). Driving experience ranged from 0 to 52 years, with 29% having over 30 years of experience. All participants were regular commuters, primarily using personal vehicles.

3.2 Procedure

After obtaining informed consent, we collected demographics via pre-survey. To ground participants' responses in a concrete representation of AV technology, we showed them a pre-recorded video simulation demonstrating how an automated vehicle operates during a journey. This video-elicitation approach [4] provided participants with a shared visual reference before discussing learning possibilities, reducing reliance on abstract imagination. The interviews were conducted remotely via video conferencing. Our interview guide included 14 core questions covering current commuting experiences, learning preferences, and perspectives on AV-based learning. Educators and automotive experts received additional domain-specific questions. All interviews were audio-recorded with consent.

3.3 Analysis

We used a general inductive thematic analysis [18]. Audio recordings were transcribed using automated tools and manually verified for accuracy. Two researchers independently read all transcripts and then collaboratively developed an initial codebook through discussion. We iteratively refined codes through multiple passes and grouped related codes into themes. Disagreements were resolved through discussion until consensus was reached. We used Atlas.ti to organize codes and examine relationships between themes. Through this process, we identified three challenges and four opportunities. We validated each theme by checking that it appeared across multiple participant groups. All interview transcripts are available upon request for research purposes.

4 Results

Our analysis identified three challenges and four opportunities for AV-based learning. We present findings organized by these themes and note how they differ from general mobile learning contexts.

4.1 Challenges

Managing Cognitive Load and Distraction. The dynamic AV environment poses attention challenges distinct from stationary learning. Eleven participants described concerns about maintaining focus. For example, P9 stated, *"I think you will be easily distracted by your surroundings because your surroundings will be constantly changing."* Unlike passive transit where passengers disengage from vehicle operation, some participants anticipated monitoring the AV. P3 noted, *"I would be looking up every now and then to see what's around me and if the car is driving well."* Physical discomfort also emerged as a concern. P8 reported, *"Sometimes I have gotten mild*

motion sickness while reading, and you're less likely to get motion sickness when you're looking out the window."

Technology Integration and Privacy. Seven participants raised concerns about system robustness and data handling in mobile environments. P28 observed, *"Implementing a conversational agent should vary based on how and where you are using it. Sitting behind a computer is very different from using a smartphone, walking in the street, or driving a car."* P30 noted the importance of balancing functionality with reliability: *"You have to make a compromise between maturity and reliability. Intuitiveness, balancing with reliability."* Data privacy also emerged as a concern, particularly regarding learning behavioral data and personal preferences being processed for adaptation. P30 stated, *"The user needs control over which data is used. You have to make transparent what you can deliver on adaptation and privacy."*

Content Design and Interdisciplinary Alignment. Four participants noted that content must suit the mobile context. P28 explained, *"Feedback should be easily visible from a distance, and there could be audio feedback to accompany it. You should limit text, as many people don't feel comfortable reading in a car."* Educators and technical experts showed different priorities. Educators emphasized pedagogical constraints. P19 noted, *"Self-driving doesn't mean the driver can be completely disengaged. It might be more feasible for passengers."* P22 added that design requires interdisciplinary consideration: *"From a design perspective, it's very interdisciplinary and varies from person to person. The key is for the technology to convey that you're in reliable hands, safe and secure in this environment, so you can focus on other tasks."*

4.2 Opportunities

Journey-Linked Contextual Learning. Participants identified the changing environment as a learning resource unavailable in traditional settings. P28 suggested, *"If a car has cameras on the outside, they could capture images at intervals, and models could analyze and respond. Preloading responses based on GPS data could be more effective than real-time generation."* P25 added, *"You could learn a new language by interpreting the road signs you pass."* P20 envisioned geography-based learning: *"The changing environment could enhance learning. If someone is interested in geography, they could learn about the landscape they're passing through."* Educators valued hands-on approaches. P18 noted, *"We could stop to explore interesting sites, engaging with nature or landmarks directly."*

Productive Time Utilization. Participants valued converting idle time into learning. P1 stated, *"It's a good way of using time that would just be wasted. You could learn a foreign language, study, or prepare for a meeting."* P30 emphasized, *"The key benefit in autonomous driving is being productive. It's hard to find silence, non-distraction environments to learn with focus. If the car supports that, it would be beneficial."*

Controlled Sensory Environment. Unlike public transit, AVs enable personalized sensory settings. P30 contrasted AVs with public transportation: *"I use public transportation a lot, but it's not a private space. There are people around, you are limited in your space. I believe*

that the more automated our vehicle becomes, the more options increase to give freedom to people." P1 emphasized the need for quiet: *"To focus, I need silence. I really need it to be quiet. I don't need any distractions."* Accessibility was also emphasized. P12 stated, *"Include multiple options. Different learning abilities like dyslexia need audio. Different physical abilities need voice controls."*

Automation-Aware Personalization. Participants valued individualized experiences enabled by AI and automation. P20 stated, *"AI could be beneficial for personalized, interactive learning. It could offer one-on-one interaction."* P30 described how AVs could adapt to individual preferences: *"People can customize or select car interiors that suit different purposes. Am I more the relaxing person in the car, am I more looking for entertainment, or am I more looking for productivity, where I would count education as a part of."* P30 also emphasized adaptive interaction: *"The design that really keeps users engaged shows progress and the interaction is intuitive and reliable."*

5 Discussion

We address each RQ and then present design recommendations based on our findings.

5.1 Addressing the Research Questions

Regarding **RQ1**, stakeholders showed distinct perspectives based on their professional backgrounds. Educators emphasized pedagogical feasibility, questioning whether drivers could truly disengage and raising concerns about accommodating diverse learning needs in constrained spaces. Technical experts focused on implementation possibilities, emphasizing context-dependent design and adaptive interfaces. Despite these differences, all groups valued converting idle commute time into productive learning. This divergence indicates that AV learning systems require interdisciplinary collaboration to balance pedagogical requirements with technical constraints. Turning now to **RQ2**, three challenges emerged: managing cognitive load and distraction, technology integration and privacy, and content design with interdisciplinary alignment. Four opportunities also emerged: journey-linked contextual learning, productive time utilization, controlled sensory environments, and automation-aware personalization. AVs differ from other mobile contexts by offering private spaces with real-time environmental data. However, participants anticipated monitoring vehicle operation [2], and motion sickness concerns [3] suggest prioritizing audio-based content. Finally, regarding **RQ3**, we propose four design recommendations below.

5.2 Design Recommendations

DR1: Adaptive Focus Management. Systems should monitor learner attention using methods such as gaze tracking and adjust content complexity and delivery pace accordingly. During high-distraction periods, the system should reduce cognitive demands or pause learning activities.

DR2: Context-Aware Content Delivery. Systems should use AV sensor data to connect learning content with the immediate environment. For language learning, systems could introduce vocabulary related to visible landmarks. For geography or history, systems could provide information about passing regions. This approach

operationalizes situated learning theory [9] by grounding abstract content in observable context.

DR3: Multi-Modal Interaction. Systems should support voice commands, touch, and gesture input, allowing learners to switch modalities based on current needs. This approach accommodates varying comfort levels with screen use and supports accessibility for diverse learners [12].

DR4: Privacy-Preserving Personalization. Adaptive systems should offer transparent control over data collection and use. Users should understand what data enables personalization and retain control over their learning profiles.

5.3 Limitations and Future Work

This study used video-elicitation to ground participant responses in a visual representation of AV technology. While this approach provides more concrete grounding than purely speculative methods, participants did not experience actual AV travel. Our participant pool, while professionally diverse, may not represent all potential AV users. Future work should validate our design recommendations through prototype testing in driving simulators, which can provide realistic AV context including motion cues. Longitudinal studies could examine learning outcomes and user experience as AV technology matures.

6 Conclusion

Through interviews with 31 commuters, including educators and automotive experts, we identified three challenges and four opportunities for AV-based learning. AV-specific affordances (real-time environmental data, private controllable spaces, and extended uninterrupted time) distinguish this context from general mobile learning. We contribute design recommendations for adaptive focus management, context-aware content delivery, multi-modal interaction, and privacy-preserving personalization. As AV technology matures, these findings provide a foundation for learning systems that transform commute time into education.

References

- [1] Gabrielle Danna, Jason Randall, and Bhindai Mahabir. 2022. Commute based learning: Integrating literature across transportation, education, and i-o psychology. *Organizational Psychology Review* 13 (10 2022), 204138662211320. doi:10.1177/20413866221132060
- [2] Remo M. A. Van der Heiden, J. Leon Kenemans, Stella F. Donker, and Christian P. Janssen. 2022. The Effect of Cognitive Load on Auditory Susceptibility During Automated Driving. *Human Factors* 64, 7 (2022), 1195–1209. doi:10.1177/0018720821998850
- [3] Cyriel Diels and Jelte E. Bos. 2016. Self-driving carsickness. *Applied Ergonomics* 53 (2016), 374–382. doi:10.1016/j.apergo.2015.09.009
- [4] S. G. Henry and M. D. Fetters. 2012. Video Elicitation Interviews: A Qualitative Research Method for Investigating Physician-Patient Interactions. *The Annals of Family Medicine* 10, 2 (March 2012), 118–125. doi:10.1370/afm.1339
- [5] Gwo Jen Hwang and Po Han Wu. 2014. Applications, impacts and trends of mobile technology-enhanced learning: a review of 2008–2012 publications in selected SSCI journals. *International Journal of Mobile Learning and Organisation* 8, 2 (2014), 83. doi:10.1504/ijmlo.2014.062346
- [6] INRIX. 2023. 2023 Global Traffic Scorecard. <https://inrix.com/scorecard/>. Accessed: January 2026.
- [7] Enkelejda Kasneci, Kathrin Seßler, Stefan Küchemann, Maria Bannert, Daryna Dementieva, Frank Fischer, Urs Gasser, Georg Groh, Stephan Günemann, Eyke Hüllermeier, et al. 2023. ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and individual differences* 103 (2023), 102274. doi:10.1016/j.lindif.2023.102274
- [8] James A. Kulik and J. D. Fletcher. 2016. Effectiveness of Intelligent Tutoring Systems: A Meta-Analytic Review. *Review of Educational Research* 86, 1 (March 2016), 42–78. doi:10.3102/0034654315581420
- [9] Jean Lave and Etienne Wenger. 1991. *Situated Learning: Legitimate Peripheral Participation*. Cambridge University Press. doi:10.1017/CBO9780511815355
- [10] Nikolas Martelaro, Jaime Teevan, and Shamsi T. Iqbal. 2019. An Exploration of Speech-Based Productivity Support in the Car. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–12. doi:10.1145/3290605.3300494
- [11] Richard E. Mayer. 2024. The Past, Present, and Future of the Cognitive Theory of Multimedia Learning. *Educational Psychology Review* 36, 1 (2024), 8. doi:10.1007/s10648-023-09842-1
- [12] Stephanie L. Moore. 2007. David H. Rose, Anne Meyer, Teaching Every Student in the Digital Age: Universal Design for Learning: Association for Supervision and Curriculum Development, 1703 N. Beauregard St., Alexandria, VA 22311-1714, Product no. 101042, \$22.95 ASCD members, \$26.95 nonmembers, ISBN-0-87120-599-8. *Educational Technology Research and Development* 55, 5 (Aug. 2007), 521–525. doi:10.1007/s11423-007-9056-3
- [13] Bastian Pfleging, Maurice Rang, and Nora Broy. 2016. Investigating user needs for non-driving-related activities during automated driving. In *Proceedings of the 15th International Conference on Mobile and Ubiquitous Multimedia*. ACM, Rovaniemi Finland, 91–99. doi:10.1145/3012709.3012735
- [14] Jan L. Plass and Shashank Pawar. 2020. Toward a taxonomy of adaptivity for learning. *Journal of Research on Technology in Education* 52, 3 (2020), 275–300. doi:10.1080/15391523.2020.1719943
- [15] Baiba Pudāne, Michal Rataj, Eric J.E. Molin, Niek Mouter, Sander van Cranenburgh, and Caspar G. Chorus. 2019. How will automated vehicles shape users' daily activities? Insights from focus groups with commuters in the Netherlands. *Transportation Research Part D: Transport and Environment* 71 (June 2019), 222–235. doi:10.1016/j.trd.2018.11.014
- [16] SAE International. 2021. *Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles*. Technical Report J3016_202104. doi:10.4271/J3016_202104
- [17] Prasanga N. Samarasinghe, Wen Zhang, and Thushara D. Abhayapala. 2016. Recent Advances in Active Noise Control Inside Automobile Cabins: Toward quieter cars. *IEEE Signal Processing Magazine* 33, 6 (2016), 61–73. doi:10.1109/MSP.2016.2601942
- [18] David R. Thomas. 2006. A General Inductive Approach for Analyzing Qualitative Evaluation Data. *American Journal of Evaluation* 27, 2 (June 2006), 237–246. doi:10.1177/1098214005283748
- [19] Kurt VanLehn. 2011. The Relative Effectiveness of Human Tutoring, Intelligent Tutoring Systems, and Other Tutoring Systems. *Educational Psychologist* 46, 4 (Oct. 2011), 197–221. doi:10.1080/00461520.2011.611369
- [20] Thiemo Wambsganss and Roman Rietsche. 2019. Towards Designing an Adaptive Argumentation Learning Tool. In *ICIS 2019 Proceedings*. Association for Information Systems. https://aisel.aisnet.org/icis2019/general_topics/general_topics/3
- [21] Christopher Wilson, Diane Gyi, Andrew Morris, Robert Bateman, and Hiroyuki Tanaka. 2022. Non-Driving Related tasks and journey types for future autonomous vehicle owners. *Transportation Research Part F: Traffic Psychology and Behaviour* 85 (2022), 150–160. doi:10.1016/j.trf.2022.01.004
- [22] World Economic Forum. 2023. The Future of Jobs Report 2023. https://www3.weforum.org/docs/WEF_Future_of_Jobs_2023.pdf
- [23] Haoran Xie, Hui-Chun Chu, Gwo-Jen Hwang, and Chun-Chieh Wang. 2019. Trends and development in technology-enhanced adaptive/personalized learning: A systematic review of journal publications from 2007 to 2017. *Computers & Education* 140 (2019), 103599. doi:10.1016/j.compedu.2019.103599

Acknowledgments

Kevin Fred Mwaita is funded by the Next-Generation EU initiative (PNRR, Mission 4, Ministerial Decree 352/2022). We thank all study participants.